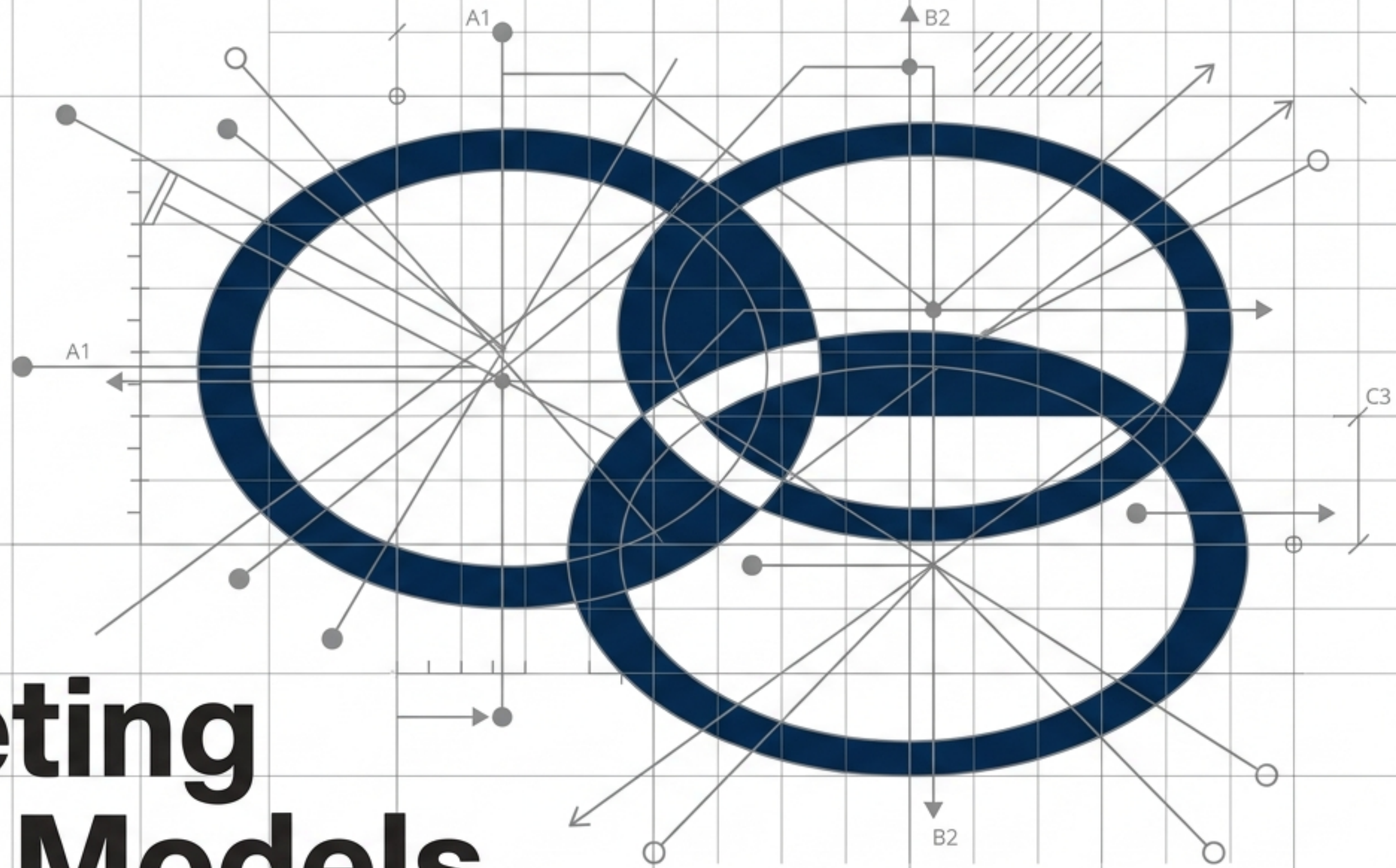




Interpreting System Models

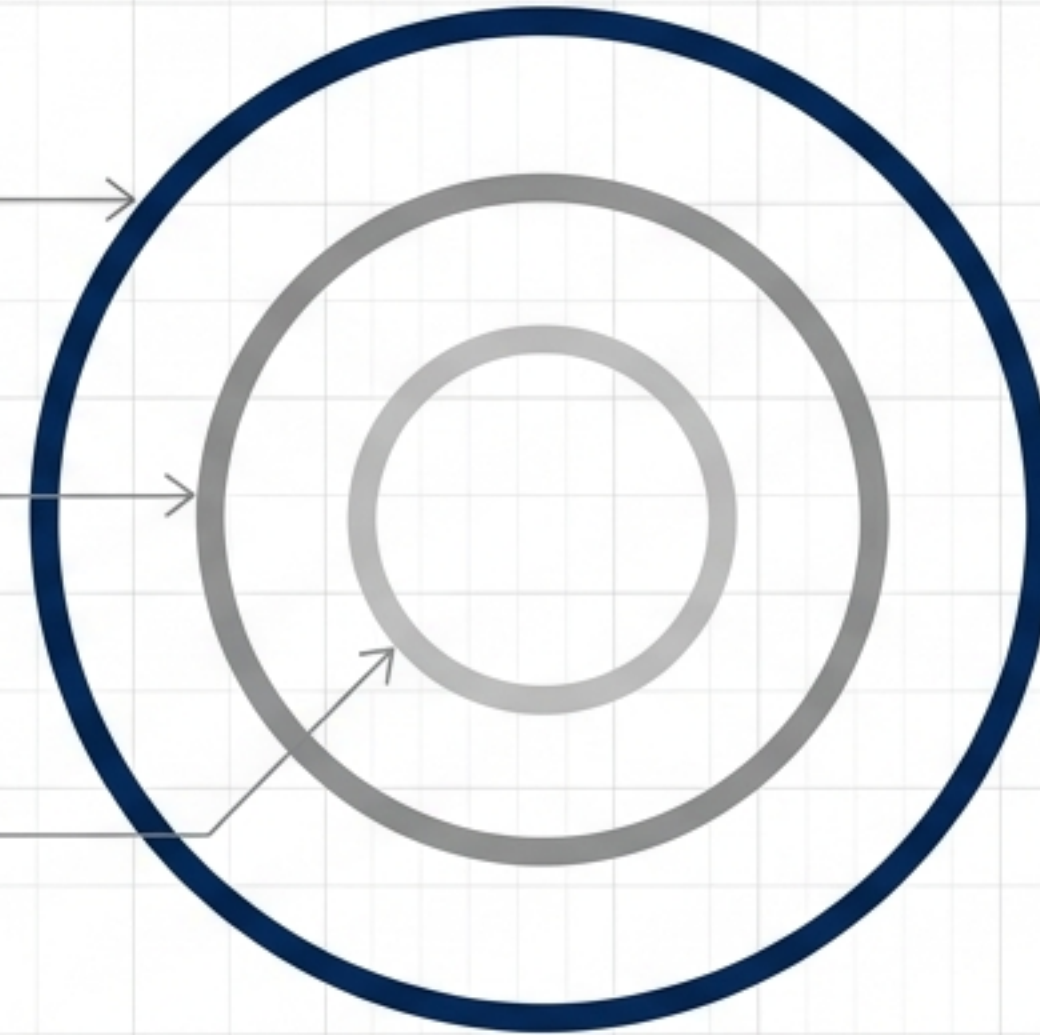
A Laboratory Guide to Interaction Modeling & Architecture

Resolving Ambiguity Across Three Scales

User Scale: Map human inputs and outputs.

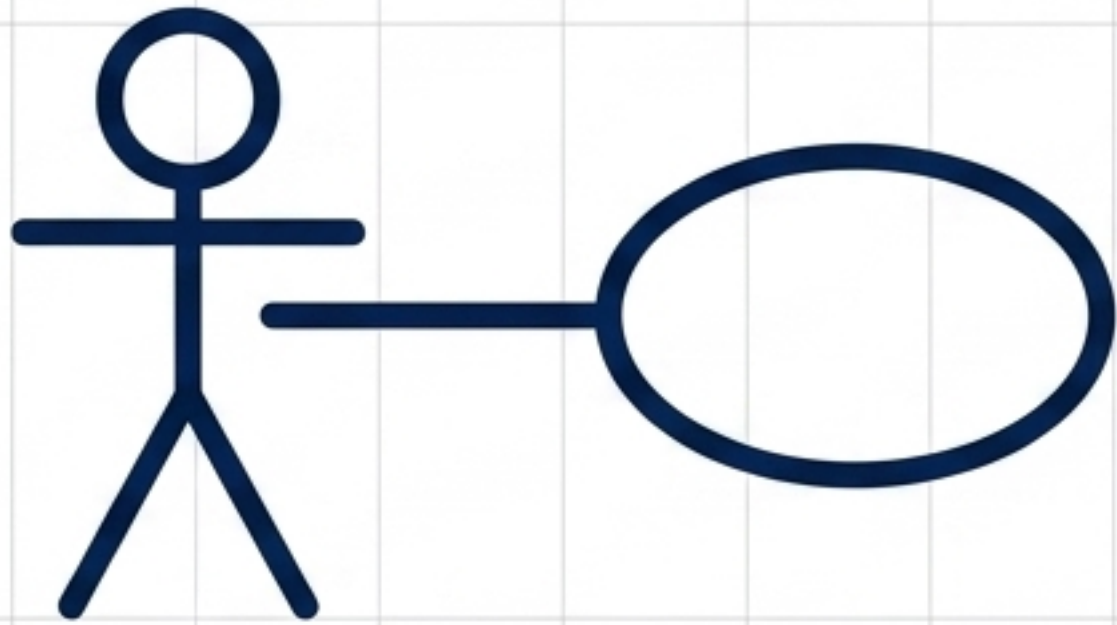
System Scale: Define communication with external environments.

Component Scale:
Examine internal dialogue and module logic.



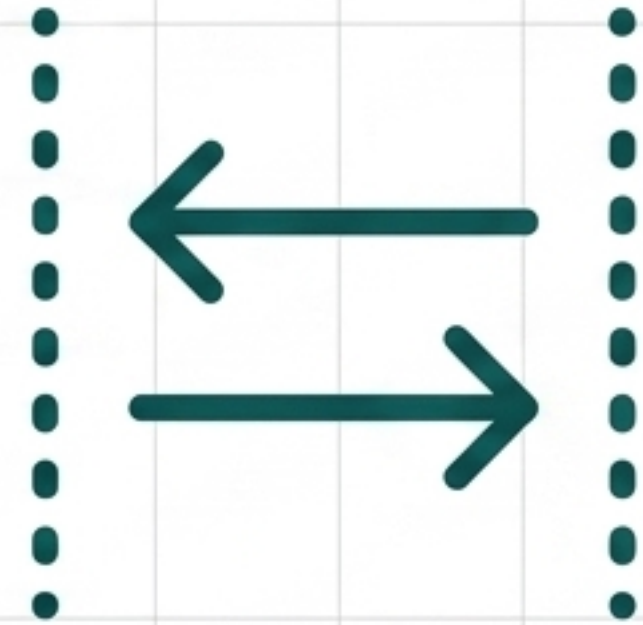
Models resolve communication ambiguity into precise architectural specifications.

The Modeling Toolkit



Use Case Diagrams

Define functional boundaries and scope. (The 'What')



Sequence Diagrams

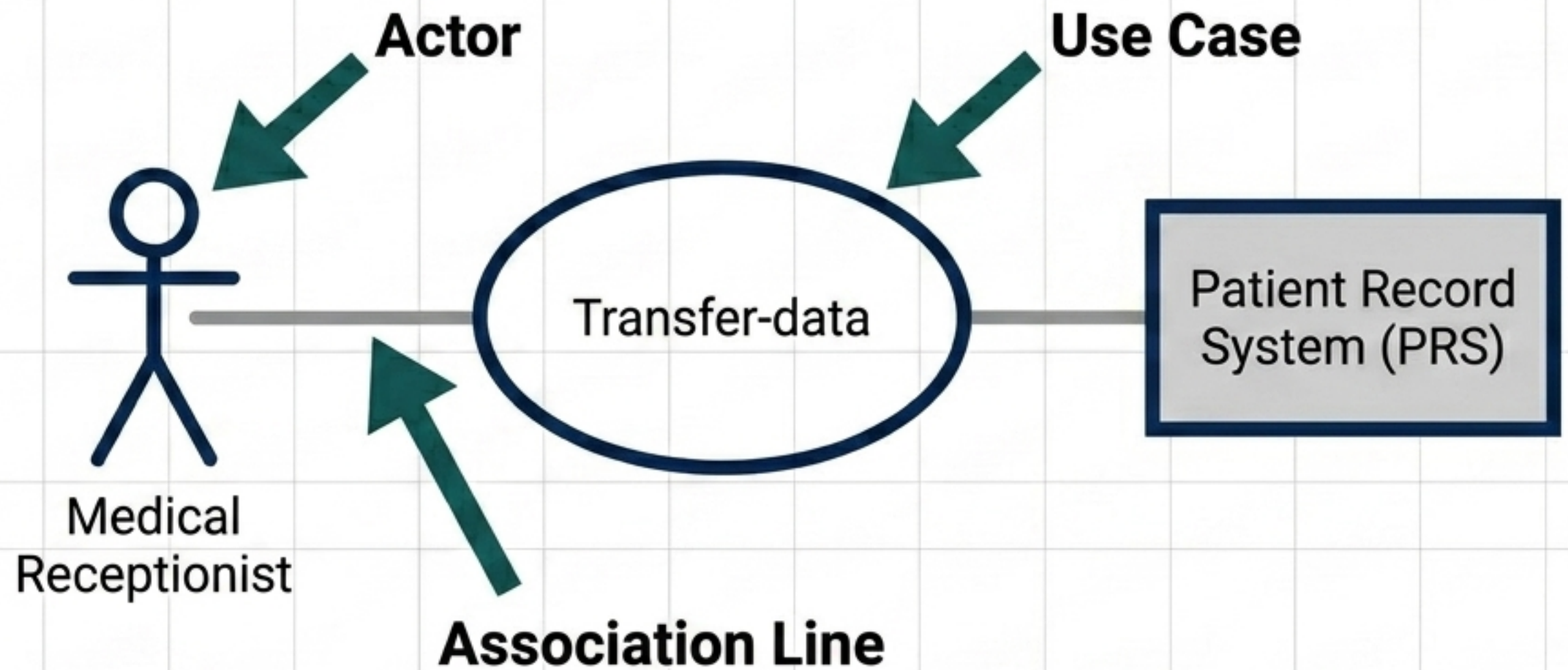
Map chronological execution. (The 'When')

Activity 1 — Defining System Scope

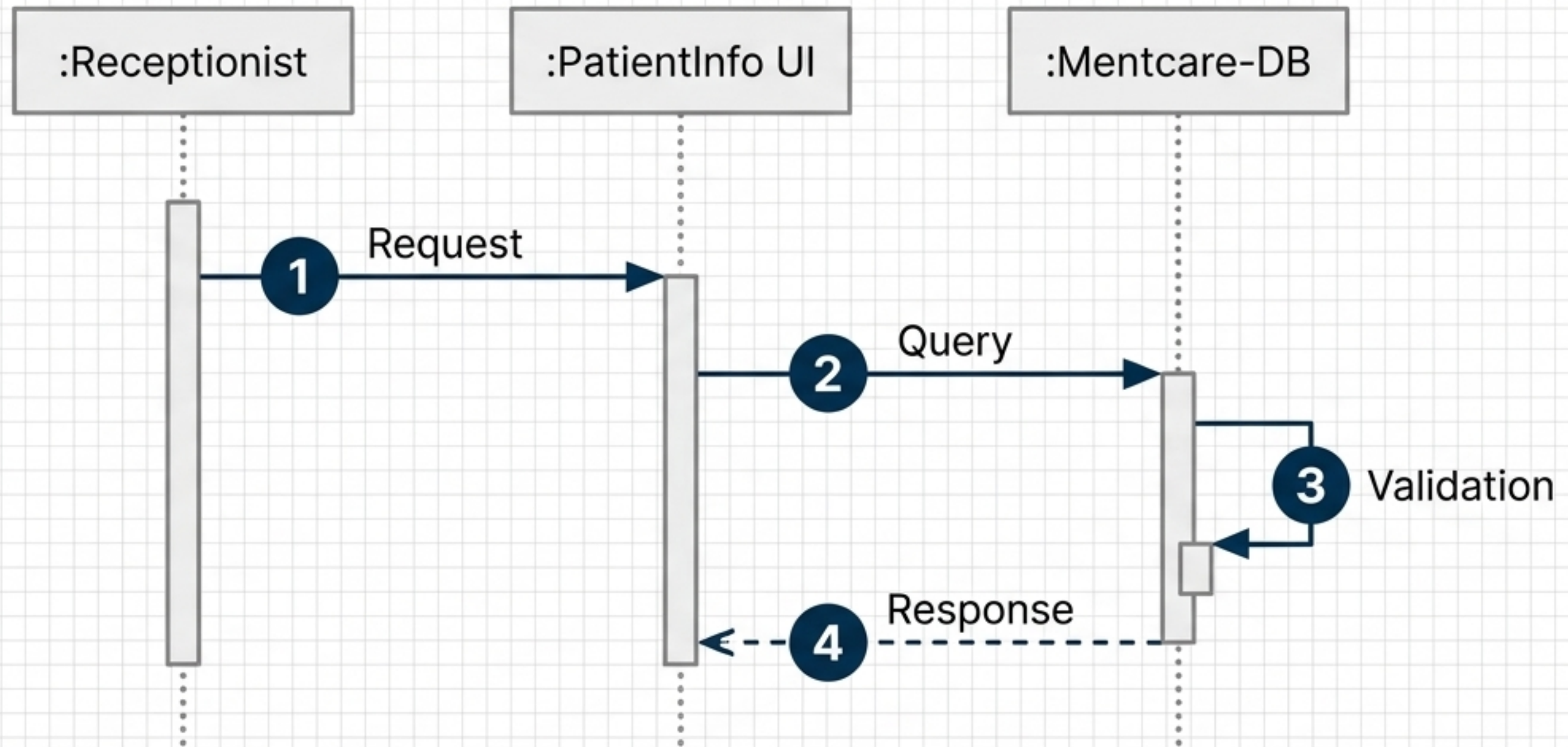
Actors: Initiators (human or external systems).

Use Cases: Discrete, observable tasks provided to actors.

Associations: Bidirectional communication links.



Activity 2 — Tracing Chronological Flow

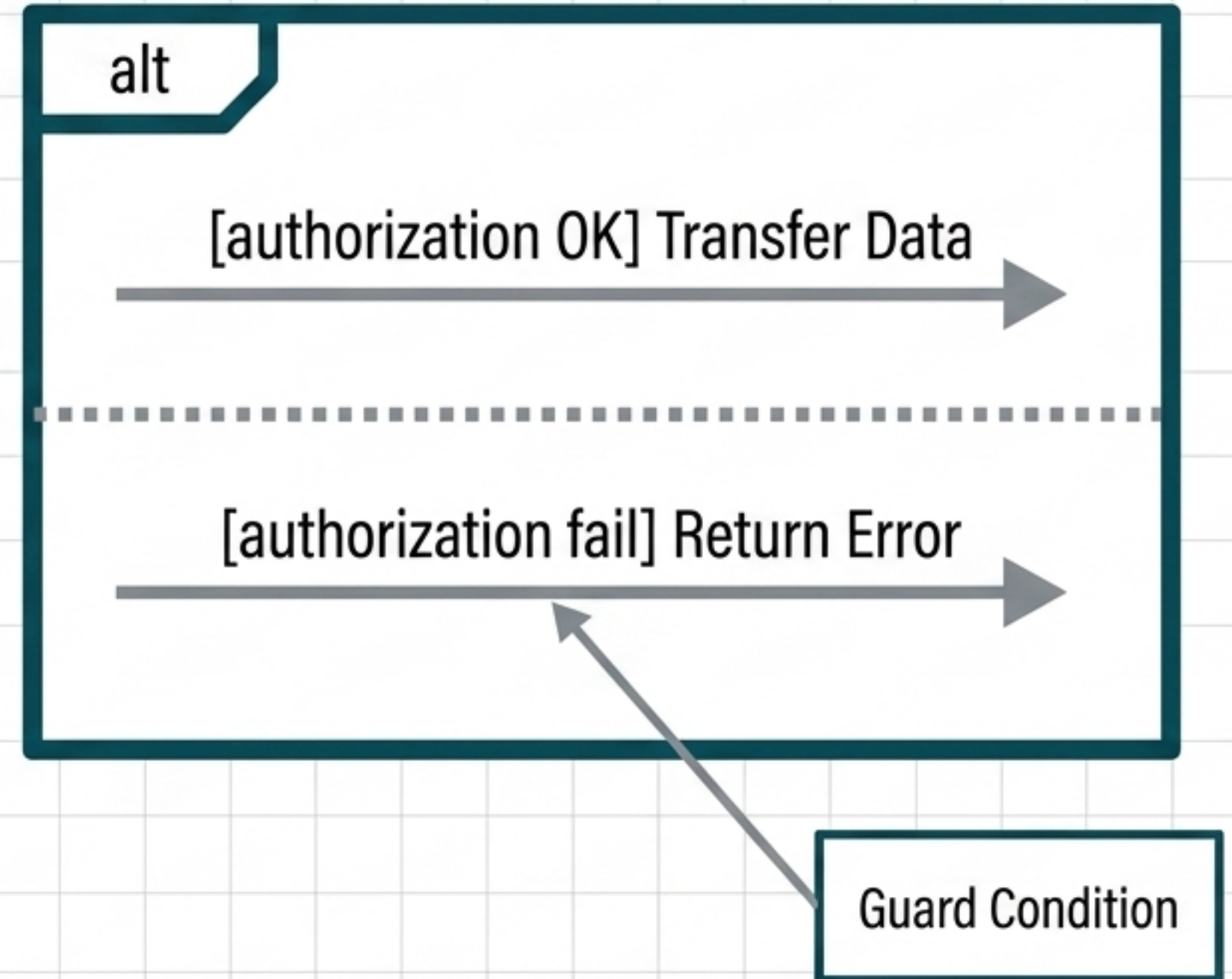


Lifelines: Vertical dotted lines (object existence)

Activation Bars: Thin rectangles (active computation)

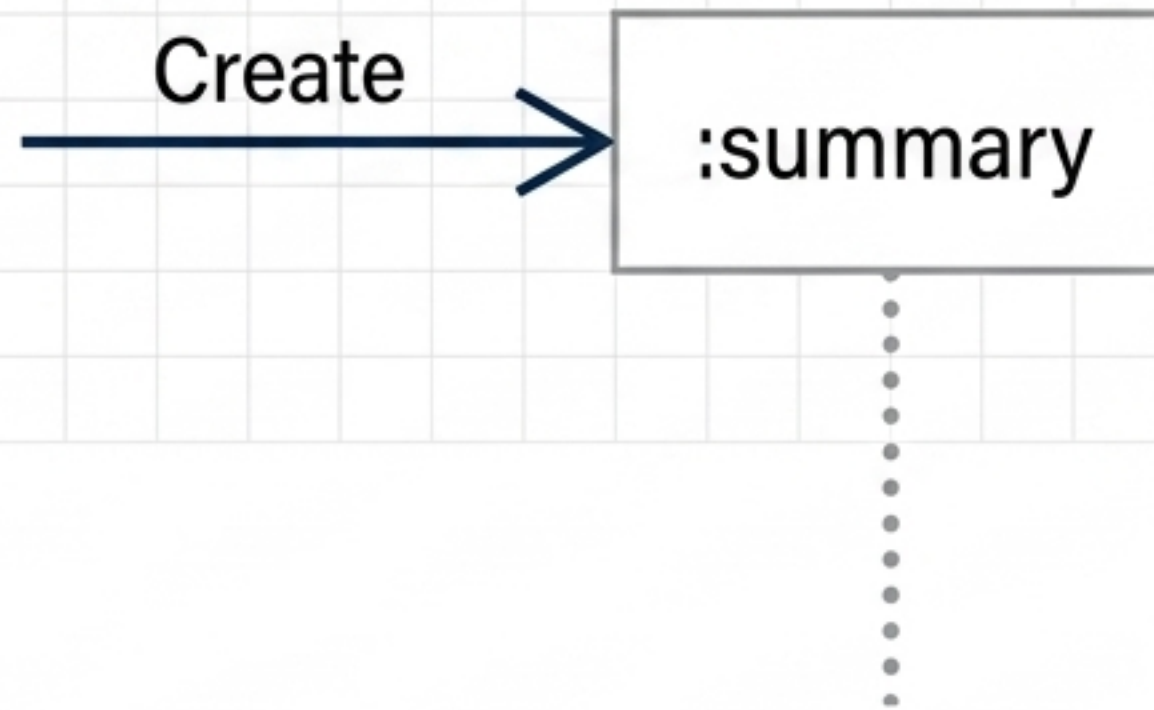
Activity 3 — Decoding Complex Logic

- **"Alt" Boxes:** Frames denoting mutually exclusive paths (e.g., personal info vs. summary).
- **Guards:** Gatekeeper conditions in brackets (e.g., [authorization OK]).
- **Purpose:** Visualizes both “happy paths” and error states simultaneously.

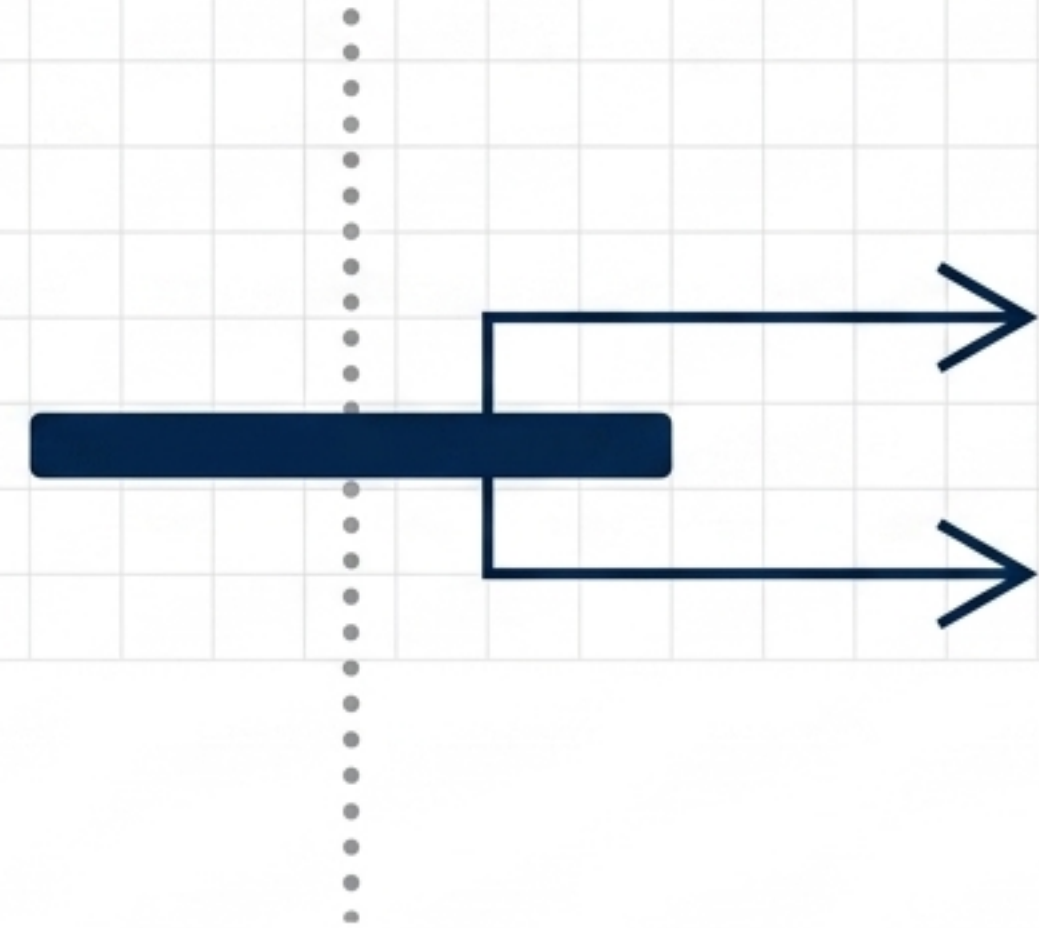


Activity 4 — Mapping Advanced Dynamics

Systems do not always wait sequentially; mature models must reflect dynamic lifecycles and parallel efficiency.



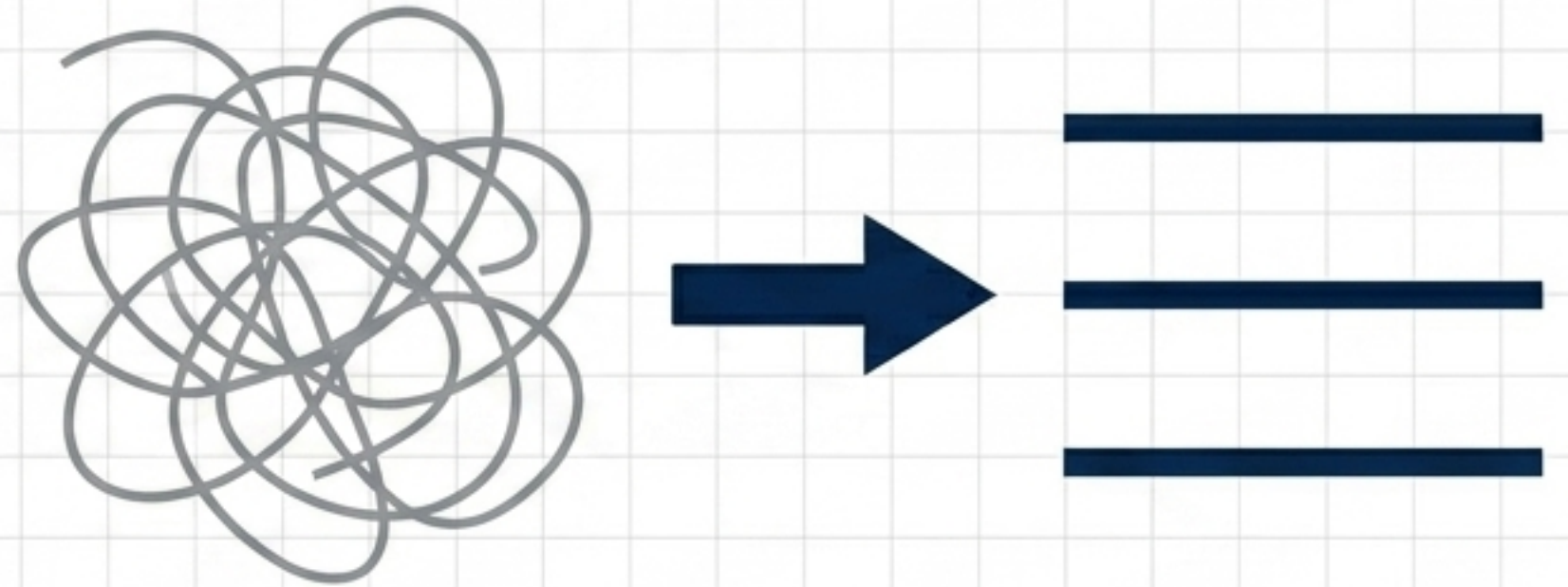
Creating objects "on the fly".



Parallel execution splitting.

Pitfalls in Model Interpretation

- **Leaky Abstractions:** Obsessing over low-level code details too early.
- **Spaghetti Logic:** Overloading a single diagram with too many paths.
- **Missing Error Paths:** Only modeling the successful execution.



Encapsulate complexity—decompose large maps into smaller, targeted specifications.

Beyond the Diagram — Tabular Metadata

Challenge: Diagrams lack granular security rules and precise input constraints.

Solution: Integrate standard tabular specifications to capture critical metadata.

Stimuli	Expected Responses	Permission Tiers
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The Model Audit Checklist



1. Who is the primary initiator?



2. What defines the system boundary?



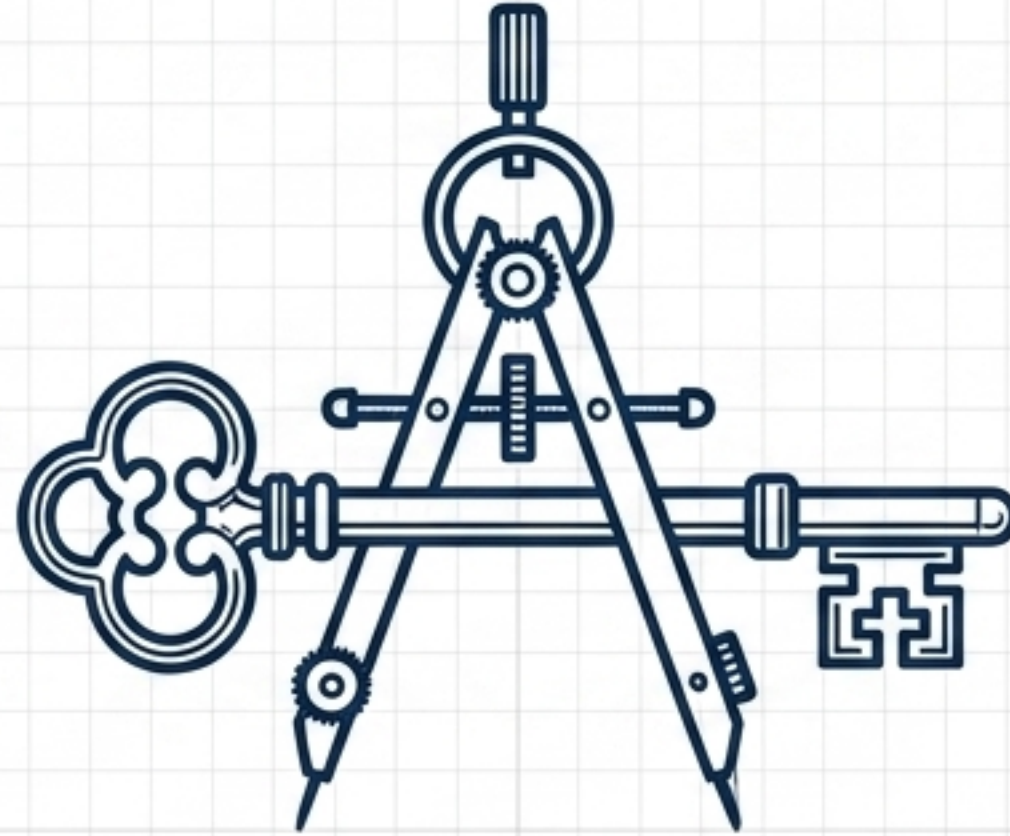
3. Is the logic exclusive ('Alt') or concurrent?



4. Are guard conditions exhaustive (errors included)?



5. Is the object lifecycle clear?



From Consumer to Gatekeeper

Models bridge the gap between abstract business requirements and technical reality.

Mastery of visual modeling validates logic before code is ever written.

You are now a gatekeeper of system integrity.